

The Wheel as an Income-Generation Strategy

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Model the Wheel as states.** Represent cash-secured-put and covered-call phases as an inventory-dependent transition system.
- **Track economic P&L.** Reconcile premiums, collateral interest, dividends, assignment settlements, transaction costs, and marked inventory through time.
- **Evaluate strategy risk.** Design strike and roll rules and backtests that measure drawdown, short-convexity exposure, assignment, and benchmark opportunity cost.

1 The Wheel Is a Policy, Not a New Payoff

The Wheel repeatedly sells cash-secured puts while out of the stock and covered calls while holding the stock. Its novelty is sequencing: assignment changes inventory, and inventory determines which option may be sold next. Each individual position retains the payoff and risk derived in L12a. Repetition does not transform short-option premium into guaranteed income.

Let $q > 0$ be the contract multiplier, usually 100 shares/contract, and consider one Wheel unit. At decision epoch $k = 0, 1, \dots$, define the inventory state

$$Z_k \in \{C, S\}, \quad (1)$$

where C denotes cash/collateral with no shares and S denotes ownership of q shares. A policy maps the current state and information $\mathcal{I}_k$ into a contract choice, position size, and management rule:

$$a_k = \pi(Z_k, \mathcal{I}_k) := (\text{option type}, K_k, T_k, \text{quantity}, \text{exit/roll rule}). \quad (2)$$

The policy must be specified before performance can be evaluated. “Sell another option” leaves strike, tenor, quote side, early exit, roll timing, and risk limits undefined.

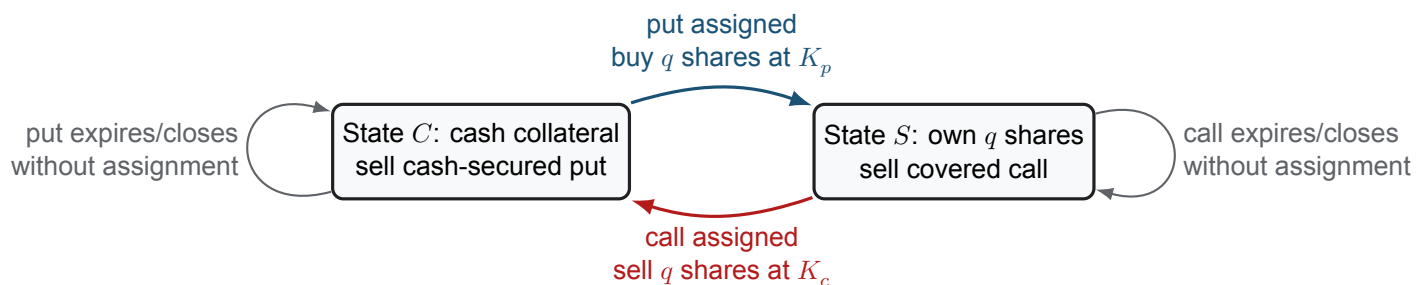


Figure 1: The Wheel as an inventory state machine. Premiums occur on every initiated option, but assignment determines whether the strategy carries cash or stock risk into the next cycle.

2 State Transitions and Inventory Conservation

In state C , sell one put at strike $K_{p,k}$ and reserve sufficient eligible collateral. If it expires or is closed without assignment, $Z_{k+1} = C$. If assigned, cash falls by $qK_{p,k}$, shares increase by q , and $Z_{k+1} = S$. In state S , sell one call at strike $K_{c,k}$ against exactly q shares. Without assignment, remain in S ; upon assignment, shares fall by q , cash rises by $qK_{c,k}$, and the next state is C .

Let $a_p \in [0, 1]$ and $a_c \in [0, 1]$ denote simplified one-cycle assignment probabilities for the put and call phases. If these probabilities were constant, the state transition matrix, with rows and columns ordered (C, S) , would be

$$\mathbf{P} = \begin{bmatrix} 1 - a_p & a_p \\ a_c & 1 - a_c \end{bmatrix}. \quad (3)$$

When $a_p + a_c > 0$, its stationary state fractions are

$$\pi_C = \frac{a_c}{a_p + a_c}, \quad \pi_S = \frac{a_p}{a_p + a_c}. \quad (4)$$

This toy Markov description clarifies that the Wheel may spend long periods in one state. In reality, assignment probabilities depend on spot, strike, tenor, volatility, dividends, rates, and the policy; states must generally include market variables and open-contract details, so [Eq. 3](#) is not a calibrated performance model.

Remark 2.1. Assignment is not the only transition mechanism

An operator may buy back, roll, exercise a long hedge, or sell shares independently. Those actions change cash and inventory even without expiration assignment. A complete state machine records every permitted action and prevents impossible states such as selling a “covered” call without the required deliverable shares.

3 The Ledger Makes or Breaks the Analysis

Let X_k be cash immediately before action k , $N_k \in \{0, q\}$ the share count, and S_k the contemporaneous stock price. Let O_k be the signed executable value of any open option position. Marked wealth is

$$W_k := X_k + N_k S_k + O_k. \quad (5)$$

A cash ledger over one management interval should reconcile

$$X_{k+1} = X_k + Q_k + I_k + D_k + A_k - C_k, \quad (6)$$

where Q_k is net option trade cash flow using bid/ask and multiplier, I_k is collateral interest net of financing, D_k is stock dividend cash flow, A_k is assignment or stock-trade settlement cash flow, and $C_k \geq 0$ collects commissions, fees, slippage, and taxes included in the chosen performance convention. Signs are cash-account signs, so put assignment produces $A_k = -qK_{p,k}$ and covered-call assignment produces $A_k = +qK_{c,k}$.

Premium should be recognized as an asset/liability exchange, not immediate economic profit. Selling an option raises cash $Q_k > 0$ while creating a short option $O_k < 0$ of approximately equal executable value. Wealth changes through subsequent repricing, decay, exercise, and costs. Counting premium at entry as profit while ignoring the open liability overstates returns.

For assigned stock, maintain at least three separate bases:

- tax/accounting basis, affected by jurisdiction-specific rules;

- strategy ledger basis, including allocated premiums and costs; and
- current market value, which determines present opportunity cost and risk.

Rolling a losing option does not erase the loss. The roll is a closing debit plus a new opening credit, and each leg must remain visible in [Eq. 6](#).

4 Where the Return Comes From—and What Is Sold

The Wheel is structurally short optionality. In state C , the short put has positive delta for the seller and negative gamma; losses accelerate as the stock falls. In state S , the covered call keeps positive delta and caps upside beyond the call strike. Premium compensates the seller for bearing volatility, jump, liquidity, and assignment risk; it is not an independent return source detached from those exposures.

One cycle's outcome is path-dependent. A sharp decline can assign stock far above its new market value and leave the strategy selling calls below its ledger basis or receiving little premium at higher strikes. A sharp rally during the covered-call phase can remove the shares and create large opportunity cost relative to buy-and-hold. Repeated small credits can therefore coexist with negatively skewed total returns.

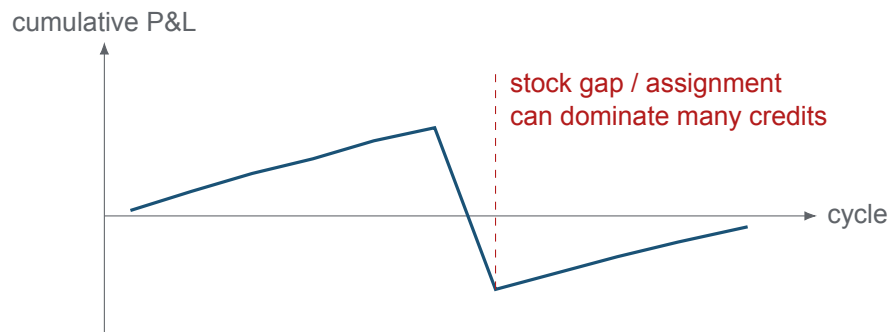


Figure 2: A schematic short-volatility return path: frequent modest credits do not bound the loss from a large underlying move. The precise sequence is policy- and market-dependent.

5 A Reproducible Wheel Policy

A testable policy specifies eligibility and action rules before observing outcomes. At minimum record:

- **Underlying eligibility:** liquidity, option volume/open interest, spread limit, price, corporate actions, and concentration caps.
- **Contract selection:** days to expiration, strike rule, quote side, minimum credit after costs, and treatment of dividends and earnings.
- **Management:** profit target, stop/risk rule, roll date, early-assignment response, and whether decisions occur intraday or at a fixed close.
- **Capital:** collateral instrument and yield, maximum utilization, reserve buffer, integer contracts, and treatment of excess cash.
- **Stock state:** covered-call strike constraints, dividend ownership, independent stock exit rules, and concentration duration.

Strike selection by delta is a sensitivity-based convention, not a literal assignment-probability guarantee. Selecting the largest premium without normalizing for width, time, volatility, and downside exposure

systematically favors riskier obligations.

COMPANION NOTEBOOK**L12b: Cash-Secured Put Phase** →

Analyze the share-acquisition phase, including collateral, premium, assignment, breakeven, and terminal downside.

COMPANION NOTEBOOK**L12b: Covered Call Phase** →

Analyze the share-sale phase, including capped upside, stock downside, assignment, and opportunity cost relative to holding shares.

6 Backtest the Process, Not Isolated Diagrams

A credible backtest uses point-in-time option chains and stock data, executable bid/ask assumptions, correct multipliers, corporate-action adjustments, dividends, collateral yield, and explicit assignment/roll logic. It must prohibit look-ahead in strike selection and avoid survivorship bias in the underlying universe. When historical chains are unavailable, model-generated quotes should be labeled synthetic and stressed for model error.

Mark the whole account daily using Eq. 5. Report annualized return only with its capital convention and alongside maximum drawdown, volatility, expected shortfall, turnover, costs, fraction of time in states C and S , assignment frequency, concentration, and benchmark-relative return. Compare with buy-and-hold stock, cash/Treasuries, and a parity-consistent alternative. A premium total alone is not a performance statistic.

Synthetic covered calls—a deep-ITM long call plus a higher-strike short call—form a call debit spread with bounded expiration loss. They can reduce initial cash outlay but introduce expiration mismatch, bid–ask, exercise, and early-assignment mechanics. They are a distinct policy variant, not automatically equivalent to deliverable shares in an American-option account.

COMPANION NOTEBOOK**L12b: Synthetic Covered Call Extension** →

Replace stock with a deep-ITM long call, derive the resulting call-spread payoff, and inspect the capital and assignment tradeoffs.

KEY TAKEAWAYS

In this lecture, we represented the Wheel as an inventory state machine and built the ledger and evaluation rules needed to test it as a complete strategy.

- **The Wheel is path-dependent.** Assignment moves the account between cash and stock states, so one option cycle cannot describe the strategy's long-run exposure.
- **Premium is paired with a liability.** Economic P&L marks the open option and stock inventory and includes funding, dividends, settlement, and costs; entry credit alone is not profit.
- **A policy needs a full backtest.** Point-in-time chains, executable prices, transition logic, drawdown, tail risk, state occupancy, and benchmark opportunity cost are required.

Next, generalize sequential decisions to exploration–exploitation problems with stochastic multi-armed bandits.

ADDITIONAL READING

- Yakov Amihud and Haim Mendelson, “[Asset Pricing and the Bid–Ask Spread](#),” *Journal of Financial Economics* 17(2), 223–249 (1986).
- Hayne E. Leland, “[Option Pricing and Replication with Transactions Costs](#),” *Journal of Finance* 40(5), 1283–1301 (1985).
- John C. Hull, *Options, Futures, and Other Derivatives*, 11th ed., Pearson (2021), chapters on option strategies, hedging, and American exercise.

Educational use and risk notice. These notes are for instruction only and do not constitute investment advice. Repeated short-option selling can incur substantial drawdowns and assignment risk.